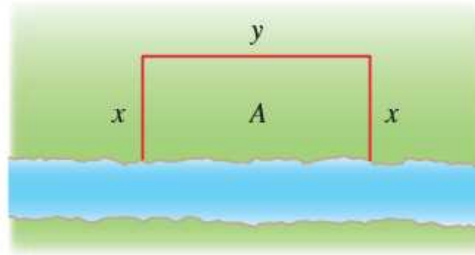




United International University
School of Science and Engineering
Department of Computer Science and Engineering
Mid Term Examination || Trimester - Spring-2026
Course Title: Calculus and Linear Algebra || Math -2183
Total Marks: 30 || Time: 1 hour 30 minutes.

Answer all the questions in order

- Q1.** (a) Consider the function $f(x) = 9x^2 - x^3$ [6]
- i. Find the intercepts of the function.
 - ii. Find the turning points and inflection points of $f(x)$.
 - iii. Find the relative maximum and minimum of $f(x)$ by using 2nd derivative test.
 - iv. Sketch the graph of $f(x)$ using the above information.
- (b) A farmer has 2400 ft of fencing and wants to fence off a rectangular field that borders a straight river. He needs no fence along the river. What are the dimensions of the field that has the largest area? [4]



- Q2.** (a) A company produces two products A and B. The profit function is $P(x, y) = 40x - 30y - 2x^2 - 2y^2 - xy$. Find the values of x and y that maximize the profit. [4]
- (b) Find the equation of the tangent plane to the surface $z = f(x, y) = x^3 + y^2 + xy$ at the point $(1, 3, f(1, 3))$ [3]
- (c) Consider the equation of a curve is given by $x^5 + y^3 - 4xy = 0$. Find $\frac{dy}{dx}$ using implicit differentiation.
- Q3.** (a) Show that $y = e^{-2x} \cos 3x$ is the solution of the differential equation $\frac{d^2y}{dx^2} + 4 \frac{dy}{dx} + 13y = 0$ [3]
- (b) An engineer is constructing a cylindrical fuel tank. The formula for the volume of a cylinder is $V = \pi r^2 h$, where r is the radius and h is the height. The engineer measures the radius as 10m with a potential error of $dr = 0.05$ and the height as 20 m with a potential error of $dh = 0.1$. What is the maximum estimated error dV in the calculated volume of the tank. [2]

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- Q4.** (a) A colony of bacteria starts with an initial population of 500. [3]
The bacteria grow at a rate at a rate proportional to the current population. If the population doubles in 4 hours, what will be the population after 10 hours?

- (b) Given that $w = xy + \ln z^2$ and $x = u^2 \cos \theta$, $y = u^2$ and $z = u^2 \sin \theta$. Find $\frac{\partial w}{\partial \theta}$ using chain rule. [3]

- Q4.** Solve the following differential equations [2]
- $\frac{dy}{dx} + 3y = 2x$, $y(0) = -1$
 - $(3x^2y + 2)dx + (3y^2x + 3)dy = 0$
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